



**SOCIAL MEDIA PERFORMANCE
ANALYSIS**
Presented By- Manpreet Kaur

JUST DO IT.

#1.

Select

```
platform,  
sum(views) as total_views,  
avg(engagement_rate) as average_engagement  
from youtube_shorts_tiktok_trends_2025  
group by platform;
```

Platform Comparison

Total views and average
engagement_rate across YouTube
Shorts vs TikTok.



Regional Hotspots

#2.

```
Select * from  
    (Select platform, country, sum(views)as total_views,  
     dense_rank()  
     over(partition by platform order by sum(views) desc) as rnk  
     from youtube_shorts_tiktok_trends_2025  
     group by platform,country)a  
where rnk<=5;
```

Top 5 countries with the highest views for Nike campaigns on each platform.



Category Performance

```
• Select
  category,
  avg(avg_watch_time_sec) as avg_watch_time_sec,
  avg(completion_rate) as avg_completion_rate
from youtube_shorts_tiktok_trends_2025
where category in ('sports', 'lifestyle', 'fashion')
group by category;
```

Average completion_rate and avg_watch_time_sec for categories like sports, lifestyle, fashion.



#4.

```
with a as(  
  Select distinct author_handle, creator_tier, creator_avg_views  
  dense_rank() over(order by creator_avg_views desc) as rnk  
  from youtube_shorts_tiktok_trends_2025  
)
```

```
Select * from a  
where rnk<=10;
```

Creator Impact

Top 10 author_handle by average views and their creator_tier.



Hashtag ROI

```
38 #5.
39 • with a as(Select
40     hashtag,
41     sum(views) as total_views,
42     dense_rank() over(order by sum(views) desc) as rnk
43     from top_hashtags
44     group by hashtag),
45
46 b as(Select
47     a.hashtag,
48     a.total_views,
49     a.rnk,
50     y.engagement_rate
51     from a join youtube_shorts_tiktok_trends_2025 as y
52     using(hashtag)
53     where rnk<=20
54 ),
```

```
56 c as(Select *,
57     row_number()
58     over(partition by hashtag order by engagement_rate) as pos,
59     count(*) over(partition by hashtag) as n
60     from b)
61
62 Select
63     hashtag,
64     total_views,
65     ROUND(AVG(engagement_rate),4) AS median_engagement_rate
66 from c
67 Where
68     pos IN (
69         FLOOR((n + 1) / 2),
70         FLOOR((n + 2) / 2)
71     )
72 GROUP BY hashtag, total_views
73 ORDER BY total_views DESC;
```

Top 20 hashtags by total views and their median engagement_rate.



Emoji Effect

Compare median engagement_per_1k between videos with and without has_emoji in titles.

```
76 • with a as (Select
77     has_emoji,
78     engagement_per_1k,
79     row_number() over(partition by has_emoji order by engagement_per_1k) as pos,
80     count(*) over (partition by has_emoji) as n
81     from youtube_shorts_tiktok_trends_2025)
82
83     Select has_emoji, avg(engagement_per_1k) as median_engagement_per_1k
84     from a
85     where pos in(
86         Floor((n+1)/2),
87         Floor((n+2)/2)
88     )
89     group by has_emoji;
```



Upload Timing

Average views and completion_rate by publish_dayofweek and upload_hour.

```
91      #7.  
92 •    Select  
93         publish_dayofweek,  
94         upload_hour,  
95         avg(views) as avg_views,  
96         Round(avg(completion_rate),4) as avg_completion_rate  
97     from youtube_shorts_tiktok_trends_2025  
98     group by publish_dayofweek,upload_hour  
99     order by publish_dayofweek;  
100
```



Trend Momentum

Median engagement_velocity and trend_duration_days by trend_type.

```
102 • with a as(Select
103     engagement_velocity,
104     trend_type,
105     row_number() over(partition by trend_type order by engagement_velocity) as pos,
106     count(*) over(partition by trend_type) as n
107 from youtube_shorts_tiktok_trends_2025),
108
109 c as (Select trend_type, avg(engagement_velocity) as median_engagement_velocity
110 from a
111 where pos in (
112     Floor((n+1)/2),
113     Floor((n+2)/2)
114 )
115 group by trend_type),
116
```

```
b as(Select
    trend_duration_days,
    trend_type,
    row_number() over(partition by trend_type order by trend_duration_days) as pos,
    count(*) over(partition by trend_type) as n
from youtube_shorts_tiktok_trends_2025),

d as(Select trend_type, avg(trend_duration_days) as median_trend_duration
from b
where pos in (
    Floor((n+1)/2),
    Floor((n+2)/2)
)
group by trend_type)
```

```
132 Select
133     c.trend_type,
134     c.median_engagement_velocity,
135     d.median_trend_duration
136 from c join d
137 using(trend_type);
```



Device Analysis

Completion rates by device_type (e.g., mobile vs tablet) and device_brand.

```
140 • Select
141 Case
142     when device_brand LIKE '%Apple%'
143     OR device_brand LIKE '%iPhone%'
144     OR device_brand LIKE '%iPad%'
145     then 'iOS'
146     else 'Android'
147     end as fixed_device_type,
148     device_brand,
149     avg(completion_rate) as avg_completion_rate
150 from youtube_shorts_tiktok_trends_2025
151 group by fixed_device_type,device_brand;
152
153 #10
```



Traffic Sources

Breakdown of traffic_source (e.g., For You Page, search, direct) with median completion_rate

```
153 #10.
154 • with a as(Select
155     traffic_source,
156     completion_rate,
157     row_number() over(partition by traffic_source order by completion_rate) as pos,
158     count(*) over(partition by traffic_source) as n
159 from youtube_shorts_tiktok_trends_2025)
160
161 Select traffic_source, round(avg(completion_rate),3) as median_completion_rate
162 from a
163 where pos in (
164     floor((n+1)/2),
165     floor((n+2)/2)
166 )
167 group by traffic_source;
168
```



Seasonal Insights

Compare views and engagement_rate across event_season (e.g., Olympics, Christmas, Summer sales)

```
168
169 #11.
170 • Select
171     event_season,
172     sum(views) as total_views,
173     round(avg(engagement_rate),4) as avg_engagement_rate
174 from youtube_shorts_tiktok_trends_2025
175 group by event_season
176 order by sum(views) desc;
177
178 #12.
179 • SELECT
180
```



Data Quality Check

Validate if engagement_total =
likes + comments + shares + saves.
Return mismatched rows.

```
179 • SELECT
180
181     engagement_total AS stored_engagement,
182
183     (
184         COALESCE(likes,0) +
185         COALESCE(comments,0) +
186         COALESCE(shares,0) +
187         COALESCE(saves,0)
188     ) AS calculated_engagement
189
190 FROM youtube_shorts_tiktok_trends_2025
191
192 WHERE engagement_total !=
193     (
194         COALESCE(likes,0) +
195         COALESCE(comments,0) +
196         COALESCE(shares,0) +
197         COALESCE(saves,0)
198     );
199
200
```



DASHBOARD VISUALIZATION & BUSINESS RECOMMENDATIONS





Platform Strategy



Deliberately increase YouTube Shorts investment to target a 50/50 platform split within 12 months, reducing over-reliance on TikTok which currently holds 60.3% of total views — a concentration risk in the event of regulatory changes or algorithm shifts.



Implement a cross-posting content framework that adapts — not just repurposes — TikTok-native content for YouTube Shorts' distinct discovery mechanics.



Build an evergreen content buffer of 8–10 videos per month to publish during historically low-performance windows (February, April, June) and stabilise the monthly view volatility.



Prioritise Evening as the primary upload window across all creators, as it generates the highest engagement across both platforms. Test Morning as a secondary slot.





Platform

TikTok

YouTube

Country

All

Category

☐ Art

☐ Automotive

☐ Beauty

☐ Comedy

Time of Day

Afternoon

Evening

Morning

Night

Month

01-01-2025



01-08-2025



Executive

Content

Creator

Market & Device

Trend

99.29K

Avg Views Per Video

21.66s

Avg Watch Time

7.54%

Engagement Rate

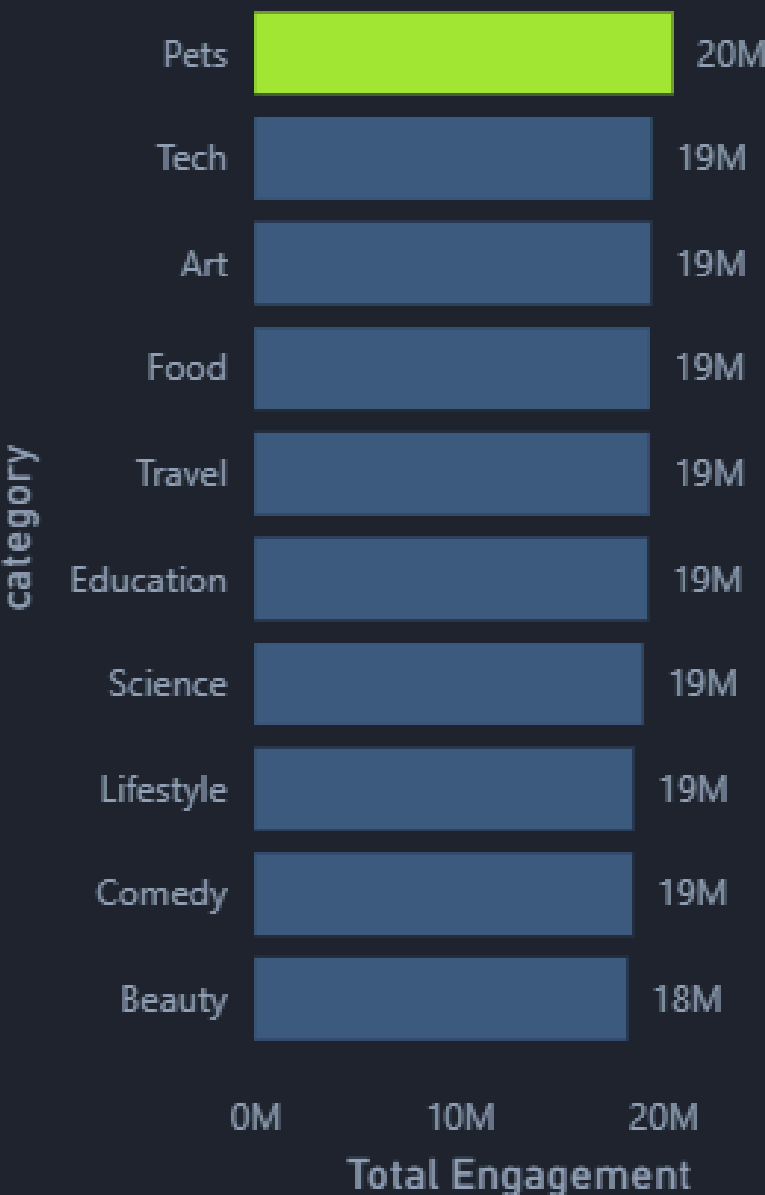
63.53%

Completion Rate

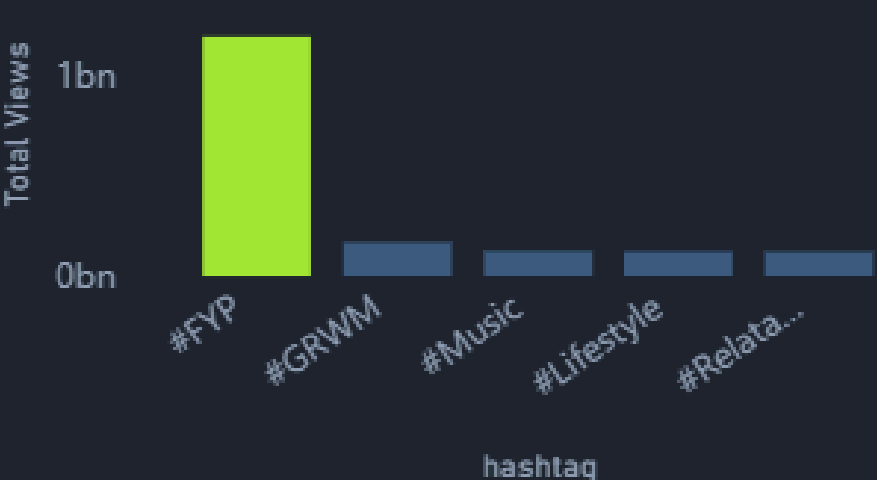
5.76%

Avg Like Rate

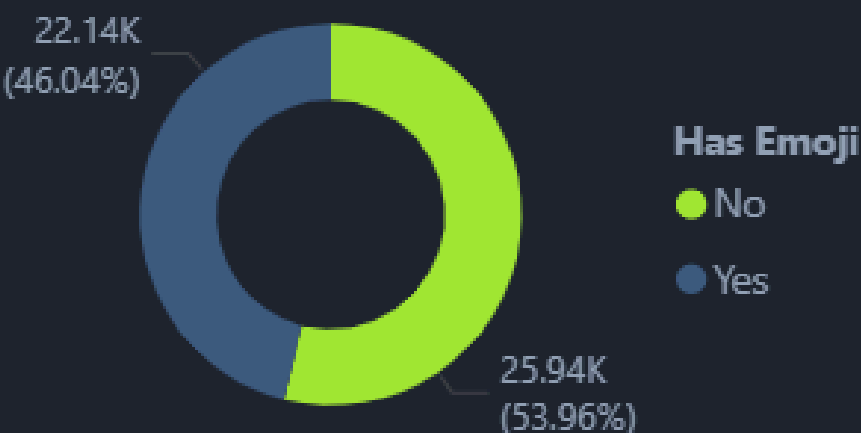
Top 10 categories



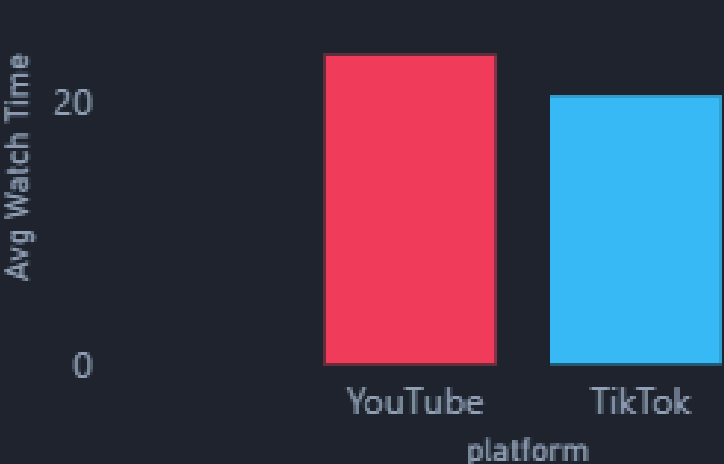
Top 5 Hashtags



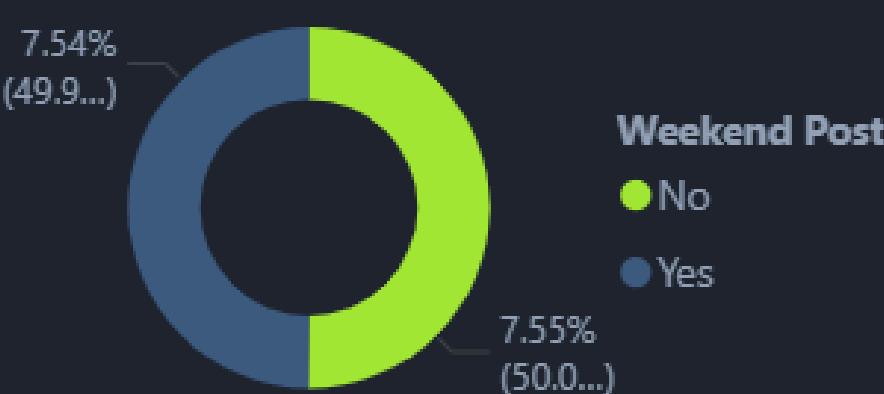
Title with emoji



Avg Watch Time by platform



Weekday vs Weekend Engagement



Content related to hashtags and emoji-based titles performs strongly, with TikTok showing slightly lower watch time than YouTube. Engagement rates remain stable overall, while categories like Art and Beauty lead in audience interaction.

01

Rebalance content categories away from Pets (currently the top category by engagement) toward Lifestyle, Fitness, and Sports — categories that deliver engagement while reinforcing Nike's aspirational brand identity.

02

Increase average watch time from 21.66 seconds to a target of 28–35 seconds by engineering stronger narrative hooks in the first 3 seconds and testing two-part content formats that drive rewatch behaviour.

03

Expand hashtag diversity beyond #FYP, which currently captures the vast majority of views while all other hashtags remain well below 0.1bn. Build a proprietary hashtag ecosystem (e.g. #NikeMove, #JustDolt2025) to own discoverability.

04

Standardise emoji usage across all content titles immediately. The data confirms that emoji-titled content accounts for 54% of posts and correlates with stronger performance — this is a zero-cost, immediate win.

05

Optimise YouTube Shorts content for deeper storytelling, where watch time is slightly higher, while keeping TikTok content focused on immediacy and trend responsiveness.

Content Strategy





Platform

TikTok

YouTube

Country

All

Category

☐ Art

☐ Automotive

☐ Beauty

☐ Comedy

Time of Day

Creator Tier

Afternoon

Macro

Evening

Mega

Morning

Micro

Night

Mid

Month

01-01-2025

01-08-2025

Executive

Content

Creator

Market & Device

Trend

648

Active Creators

7.59%

Creator Engagement

102.5K

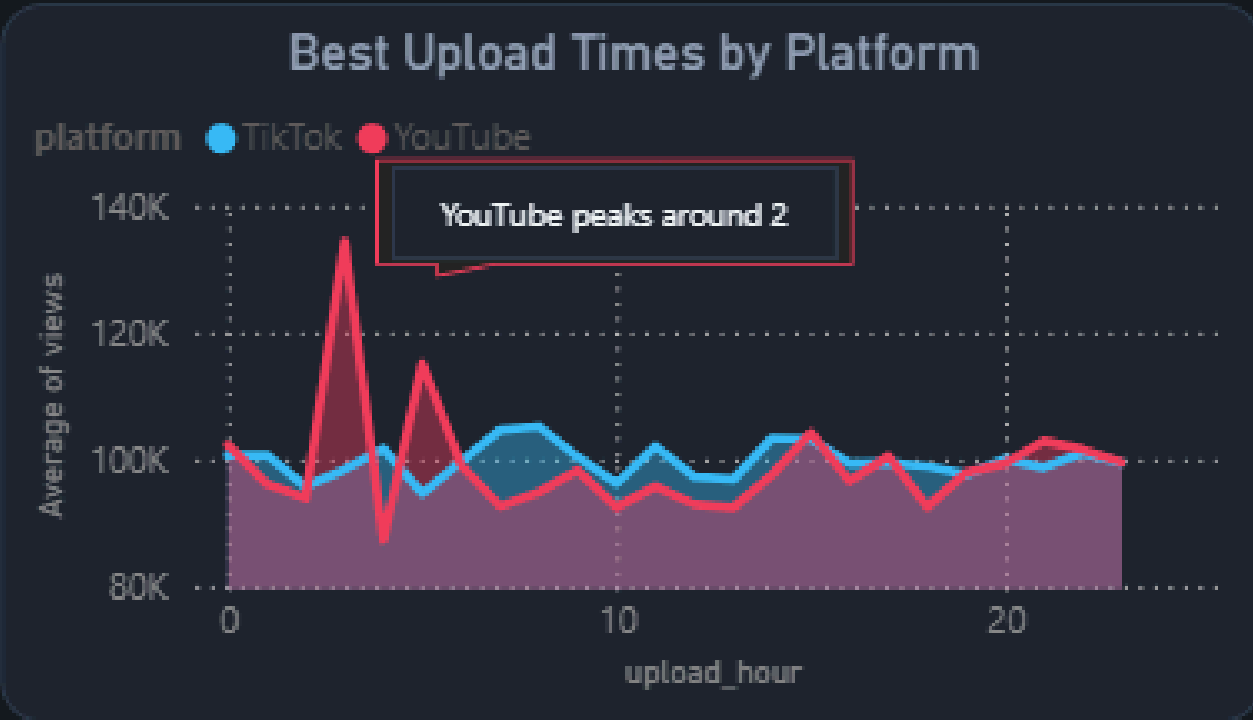
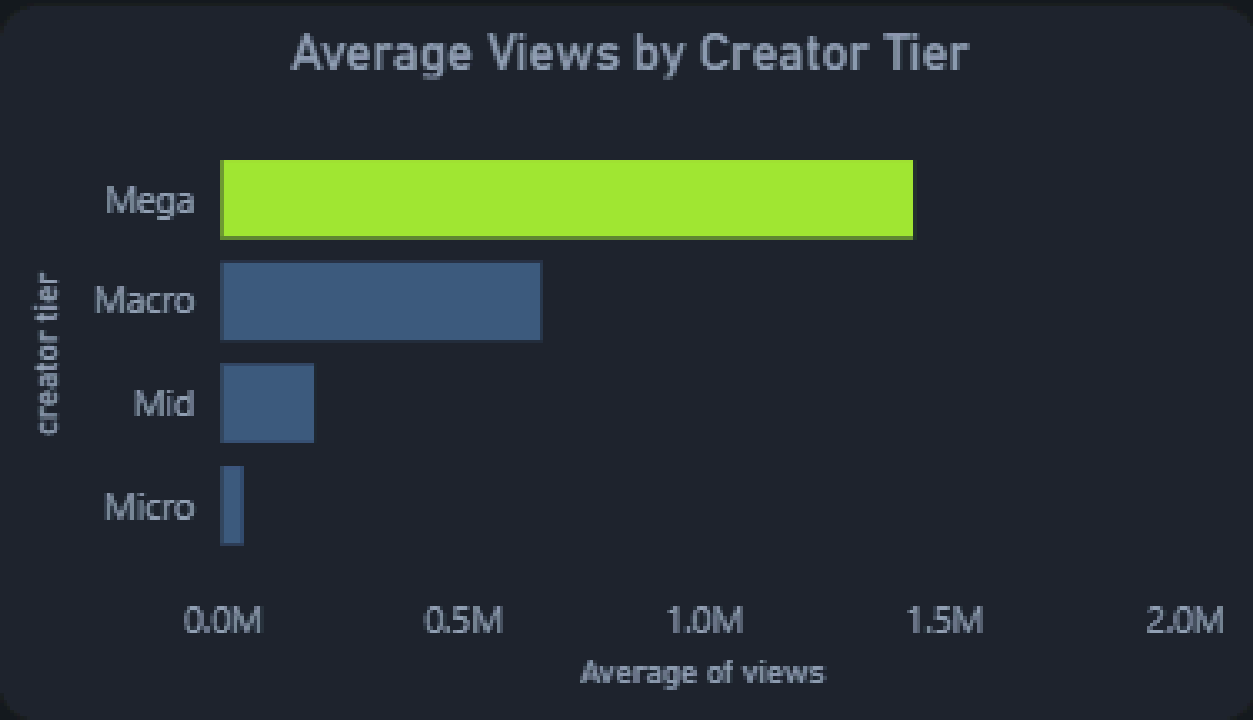
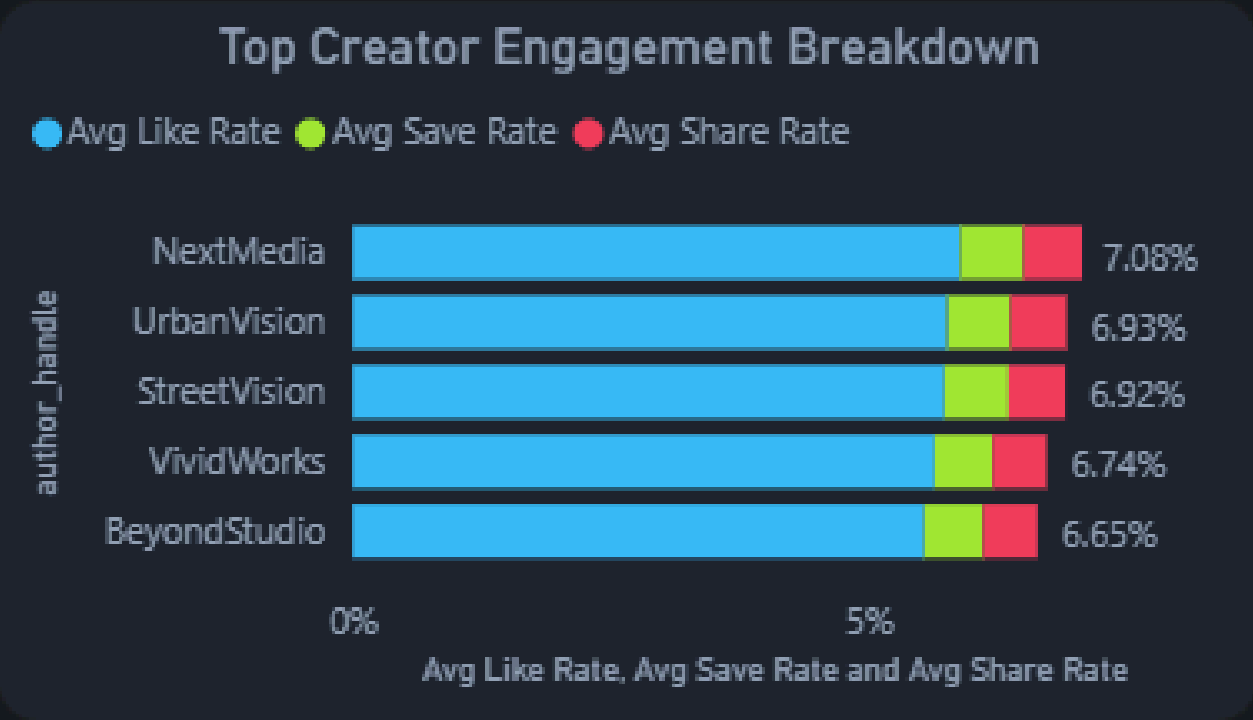
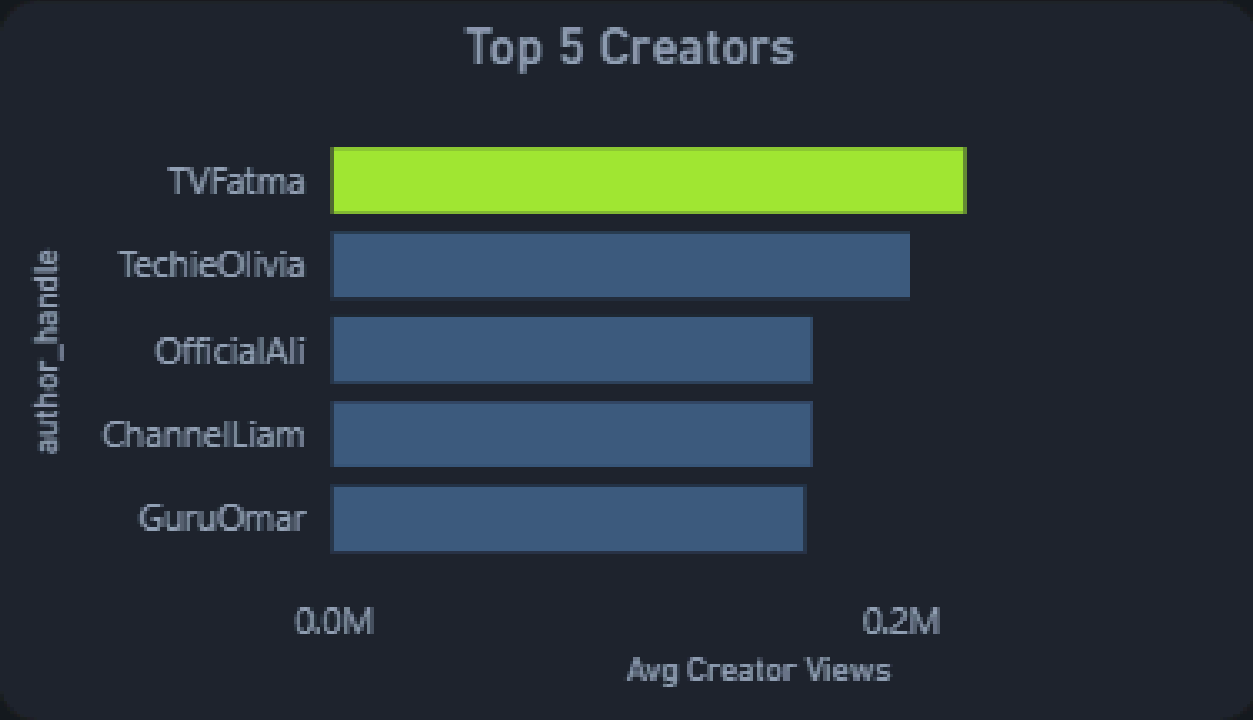
Avg Creator Views

18M

Top Creator Views

0.56%

Avg Share Rate



Top creators such as TVFatma and TechieOlivia drive the highest creator views, while Mega-tier creators significantly outperform other creator tiers in average views. YouTube shows sharper upload-time spikes, whereas TikTok maintains more consistent engagement across the day.

01

Restructure the creator programme into a dual-track model: a Reach Track (Mega and Macro creators for volume) and an Engagement Track (Mid and Micro creators for community depth and conversion intent). Fund and evaluate each track independently.

02

Activate Micro and Mid-tier creators more aggressively in niche categories — Running, Training, Sport Performance — where community trust drives purchase consideration more than raw view counts.

03

Negotiate performance-linked creator contracts that tie renewal to engagement velocity and share rate, not view volume alone, to reduce spend on high-reach but low-quality impressions.

04

Urgently address the 0.56% average share rate — the weakest metric across the entire dataset. Brief all creators specifically on shareability triggers: emotional peaks, surprise endings, challenge formats, and duet-ready content structures.

05

On YouTube, align creator upload schedules to the Hour 2 peak performance window. On TikTok, leverage the platform's more consistent engagement profile to distribute content more evenly throughout the day.

Creator Ecosystem





Platform

TikTok

YouTube

Country

All

Category

- ☐ Art
- ☐ Automotive
- ☐ Beauty
- ☐ Comedy

Time of Day

Afternoon

Evening

Morning

Night

Month

01-01-2025



01-08-2025



Executive

Content

Creator

Market & Device

Trend

Ke

Top Country

168M

Top Country Views

63.53%

Avg Completion

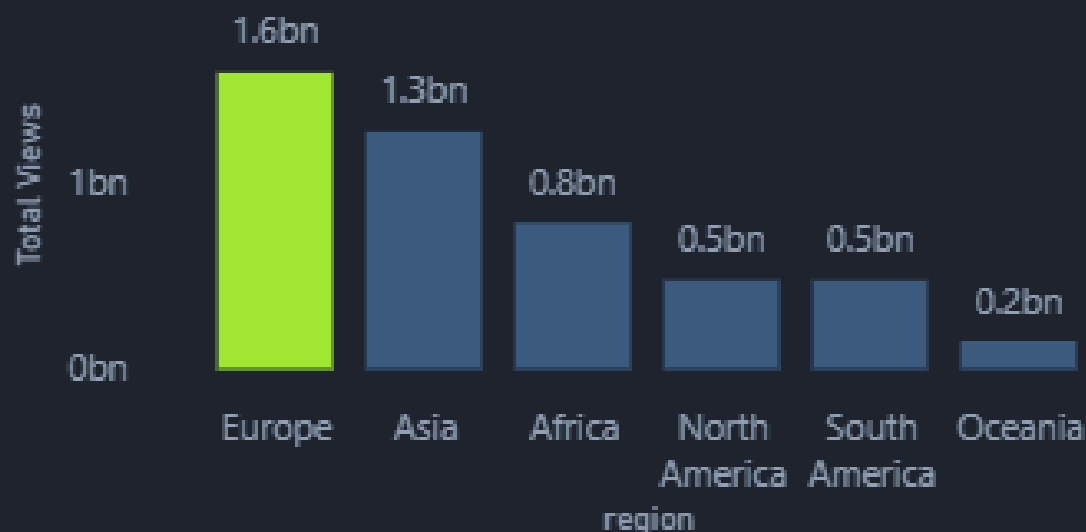
21.66s

Avg Watch Time

Android

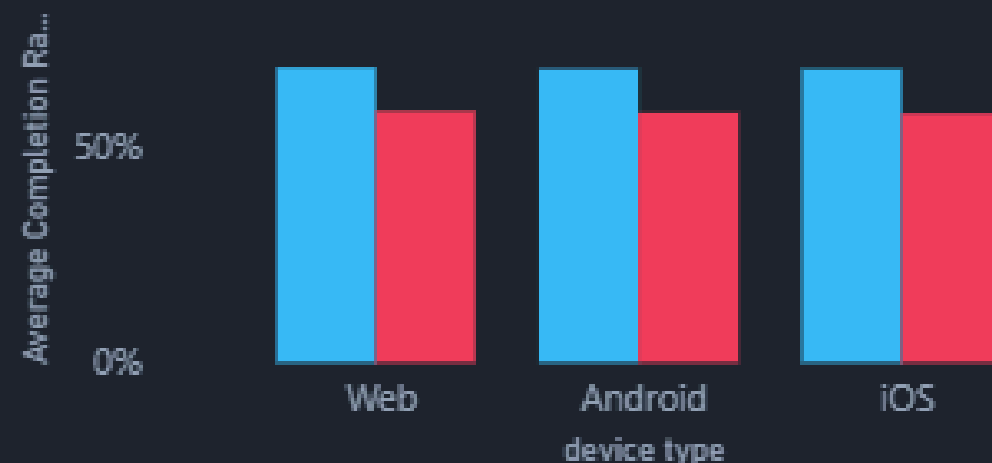
Most Used Device

Total Views by Region



Device Type Completion Comparison

platform ● TikTok ● YouTube



Device Brand

- ☐ Desktop
- ☐ Huawei
- ☐ iPhone
- ☐ Oppo
- ☐ Other
- ☐ Pixel
- ☐ Samsung
- ☐ Vivo
- ☐ Xiaomi

Total Engagement by traffic_source



Country Performance Overview



Europe and Asia dominate global viewership, while Android is clearly the most-used device. TikTok consistently achieves higher completion rates across device types, and the "For You" traffic source drives the majority of engagement.

01

Launch a dedicated North America content revival programme featuring US and Canadian athletes, cities, and subcultures. North America generates only 0.5bn views on par with South America — which is a critical underperformance for Nike's home market.

02

Treat Asia as a high-priority growth region. At 1.3bn views, Asia is near parity with Europe (1.6bn) but likely holds a significantly larger addressable audience. Commission separate, localised content strategies for India, Southeast Asia, and Japan.

03

Develop a Search Optimisation Strategy for both TikTok and YouTube Shorts including optimised titles, captions, and keywords — to grow the Search traffic source (currently 57M engagements) and reduce dependency on the For You Page (161M).

04

Build a Following and Subscription growth playbook. The Following traffic source drives 40M engagements and represents Nike's highest-intent audience. Invest in content formats specifically designed to convert casual viewers into long-term followers.

05

Implement an Android-first creative quality assurance process. Android is the most-used device in Nike's audience, yet most brands produce and review content on Apple hardware — creating a quality gap for the majority of viewers.

Market & Device Intelligence





Platform

TikTok

YouTube

Country

All

Category

- ☐ Art
- ☐ Automotive
- ☐ Beauty
- ☐ Comedy

Time of Day

Afternoon

Evening

Morning

Night

Month

01-01-2025

01-08-2025

Executive

Content

Creator

Market & Device

Trend

14

Avg Trend Duration

13.89K

Avg Eng Velocity

Summer

Top Season

Short

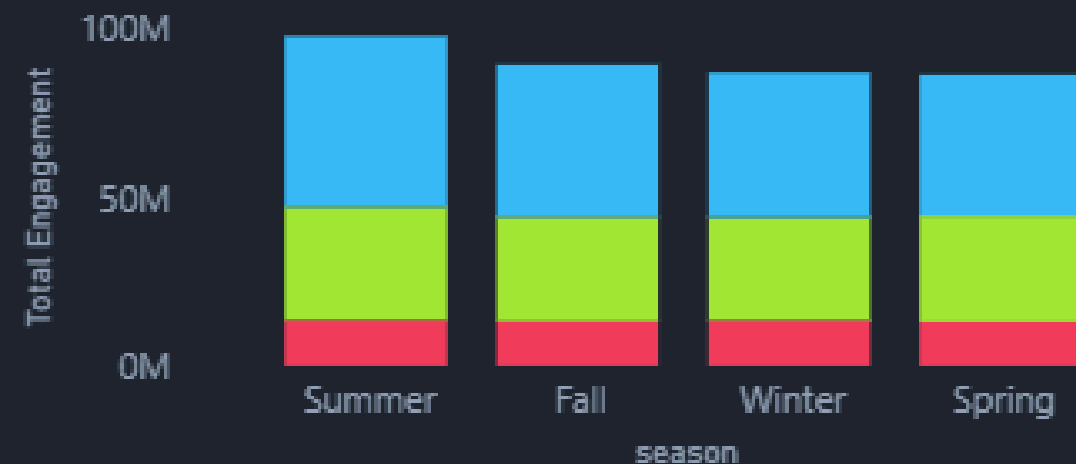
Top Trend

Trend Type

All

Engagement by Trend Type

trend_type ● Evergreen ● Medium ● Short



Seasonal Performance Pattern

trend_type ● Evergreen ● Medium ● Short

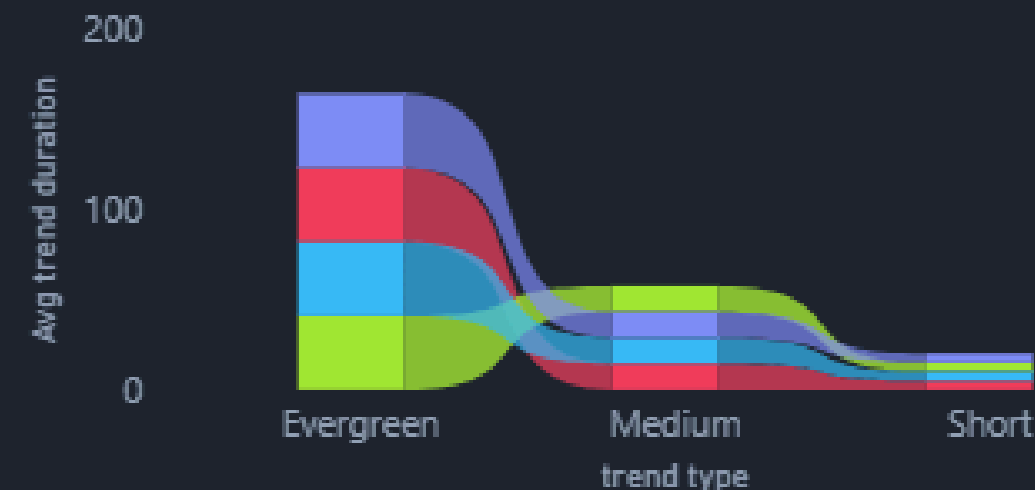


event_season	Evergreen	Medium	Short
BackToSchool	7.55%	7.44%	7.56%
HolidaySeason	7.60%	7.61%	7.33%
Ramadan	8.16%	7.36%	7.52%
Regular	7.50%	7.59%	7.55%
SummerBreak	7.55%	7.47%	7.58%

BackToSchool	7.55%	7.44%	7.56%
HolidaySeason	7.60%	7.61%	7.33%
Ramadan	8.16%	7.36%	7.52%
Regular	7.50%	7.59%	7.55%
SummerBreak	7.55%	7.47%	7.58%

Trend Duration vs Engagement

trend_label ● declining ● rising ● seasonal ● stable



Short and Medium trends generate the strongest engagement across seasons, with Summer emerging as the top-performing season overall. Trend engagement remains relatively stable across durations.

01

Build a Rapid Content Unit (RCU): a small, dedicated team of 3–5 creators and one editor with pre-approved brand assets and a 48-hour production-to-publish mandate for Short trends. The average trend lifecycle is just 14 days — speed is a direct competitive advantage.

02

Develop a formal Seasonal Content Calendar that front-loads Summer production (the top-performing season) and designates Ramadan, Back-to-School, and Holiday Season as priority campaign windows with dedicated briefs and budgets.

03

Refresh Evergreen content formats every 60–90 days with updated visual treatments to reactivate algorithm interest, as Evergreen view trajectories are currently on a declining path despite their long duration.

04

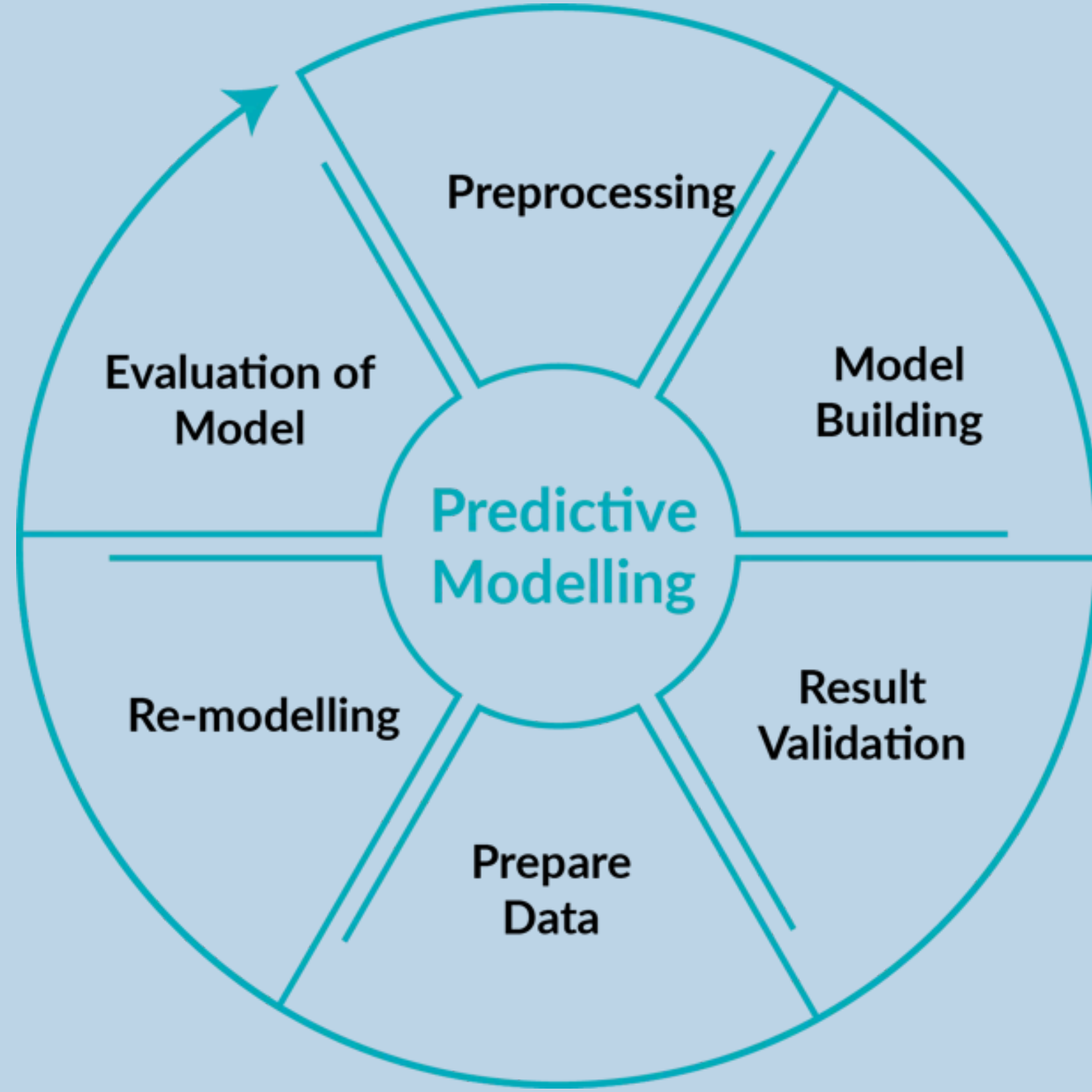
Develop a dedicated Ramadan 2026 campaign, beginning pre-production in November 2025. Ramadan delivers the single highest engagement rate in the dataset (8.16% for Evergreen content) and is an underleveraged cultural window — particularly given Nike's scale across Europe and Asia.

05

Set Short trend participation as a creator KPI: target that 40% of creator output is tied to trending formats within 72 hours of a trend surpassing significant engagement velocity.

Trend & Seasonal Strategy



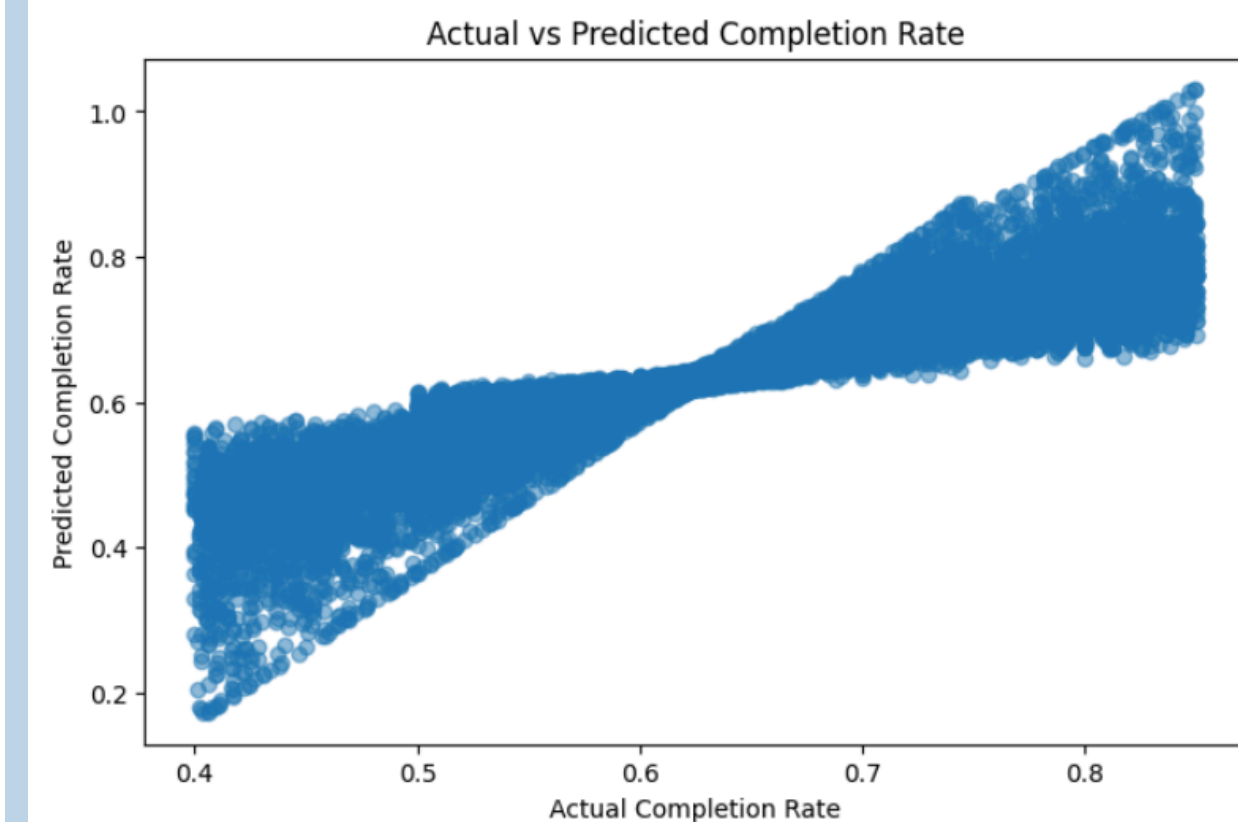
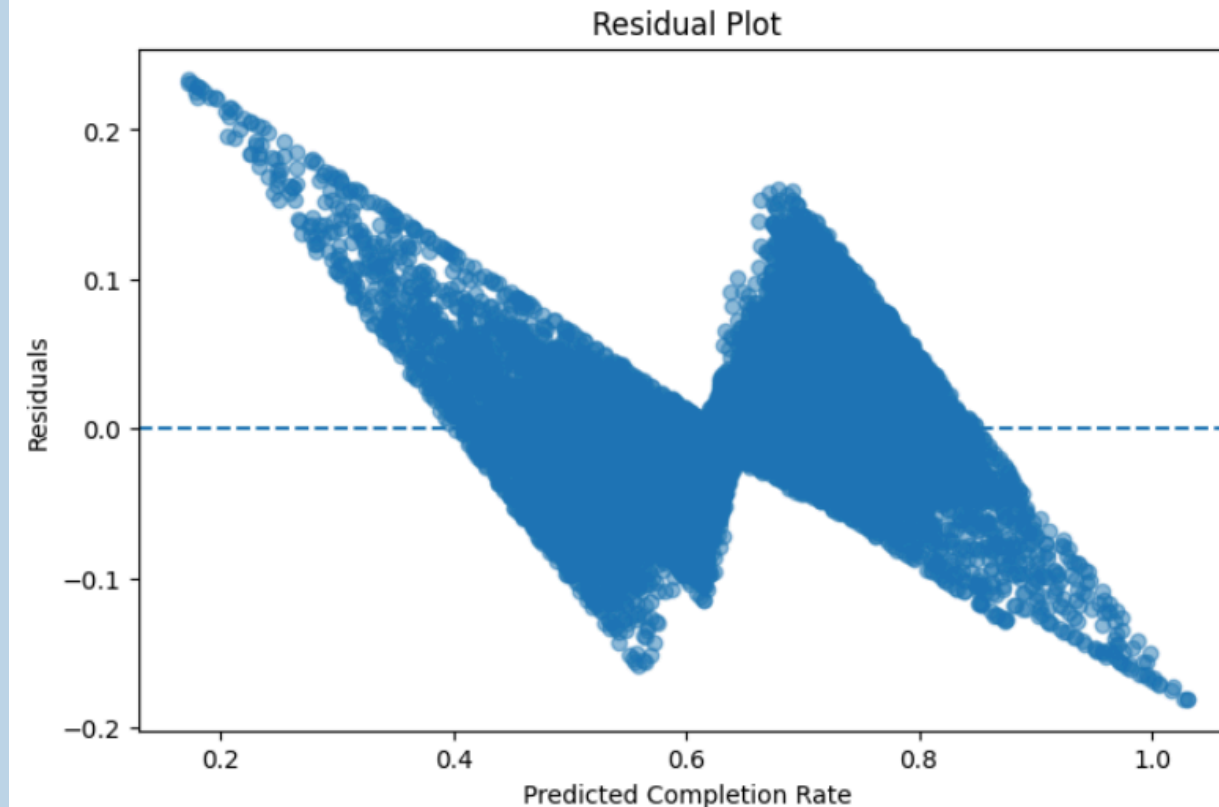
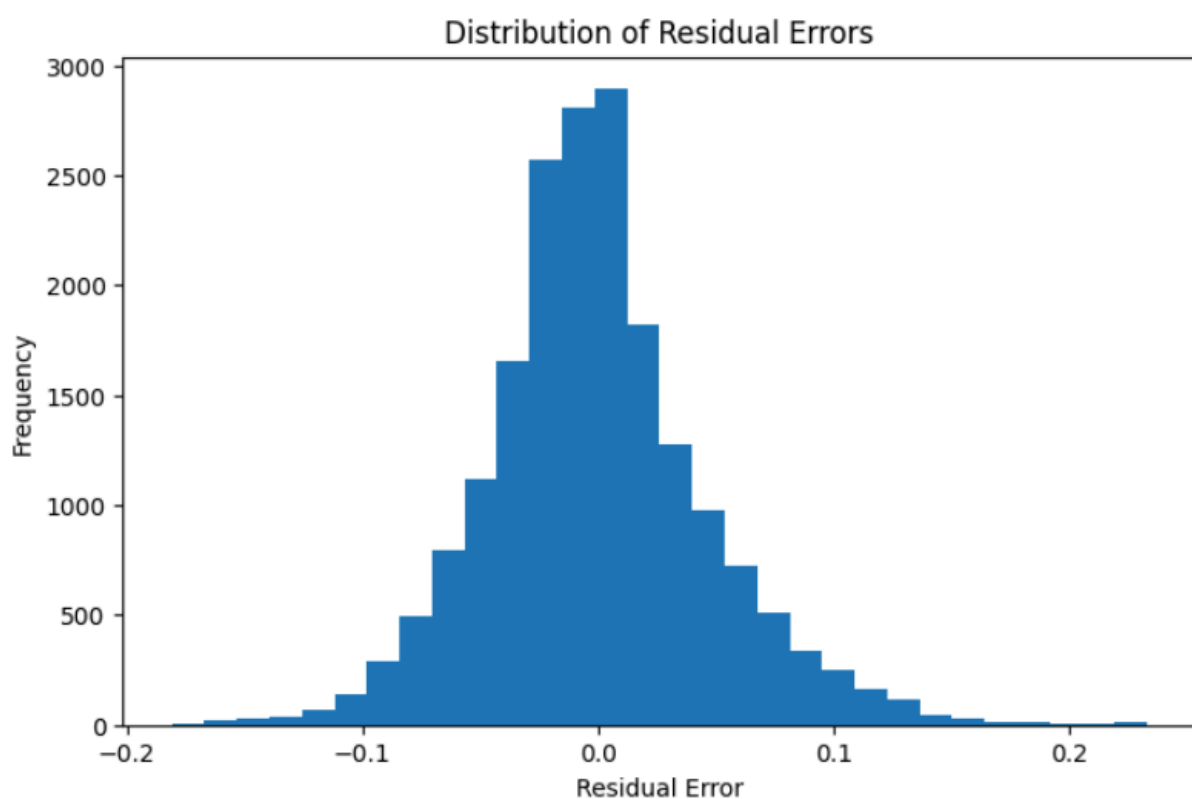


PREDICTIVE MODELING AND PERFORMANCE EVALUATION



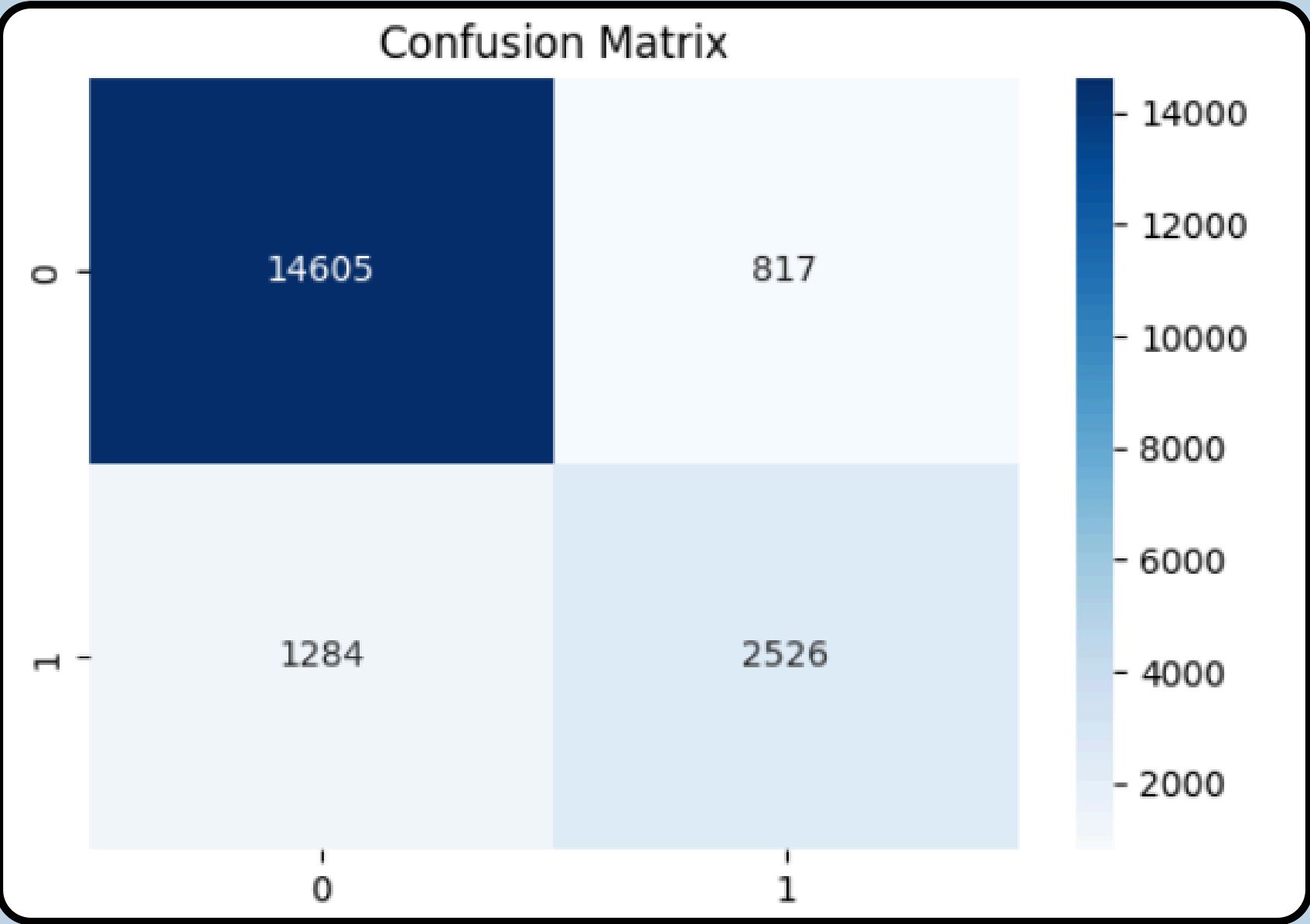
Completion Rate Prediction Using Linear Regression

A Linear Regression model was developed to predict the completion rate of reels based on various engagement and content-related features. The model analyzes the relationship between independent variables such as likes, comments, shares, saves, upload timing, and content characteristics with the target variable, completion rate. This predictive approach helps identify the factors that contribute most to audience retention and content performance.



Virality Classification Using Logistic Regression

A Logistic Regression model was implemented to classify whether a reel is likely to become viral or not. The model uses engagement metrics and content attributes to estimate the probability of virality based on historical data patterns. By transforming the output into binary categories (viral/non-viral), the model supports data-driven decision-making for optimizing content strategy and maximizing audience reach.



	precision	recall	f1-score	support
0	0.92	0.95	0.93	15422
1	0.76	0.66	0.71	3810
accuracy			0.89	19232
macro avg	0.84	0.81	0.82	19232
weighted avg	0.89	0.89	0.89	19232



**THANK
YOU**

